

Introduction to the Theory of Computation

Lecture 2. Big O Notations

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Why do we need big O notation?

- **In computer science:** to analyze the running time and space of algorithms
- **In mathematics:** to control lower-order terms without getting your hands dirty

Big O notation, also known as Landau asymptotic notation, is named after the German number theorist Edmund Landau (1877 – 1938).

Definition 1 (Big O notation)

Let $f, g : \mathbb{R} \rightarrow \mathbb{R}$. We write $f(x) = O(g(x))$ if

$$(\exists c > 0)(\exists N)(\forall x \geq N)(|f(x)| \leq c \cdot |g(x)|).$$

For example,

- $T(n) = 6n^4 - 2n^3 + 5 = O(n^4)$
- $1 + 2 + \dots + n = O(n^2)$
- $1 + 2 + \dots + n = \frac{1}{2}n^2 + O(n)$
- $1^2 + 2^2 + \dots + n^2 = \frac{1}{3}n^3 + O(n^2)$
- $\frac{1}{1} + \frac{1}{2} + \dots + \frac{1}{n} = \ln n + O(1)$
- $\frac{1}{1^2} + \frac{1}{2^2} + \dots + \frac{1}{n^2} = O(1)$

Proof: $T(n) = 6n^4 - 2n^3 + 5 = O(n^4)$

Recall

$f(n) = O(g(n))$ if there exist constants $c > 0$ and N such that

$|f(n)| \leq c \cdot |g(n)|$ for all $n \geq N$.

Note that

$$\begin{aligned} |6n^4 - 2n^3 + 5| &\leq 6n^4 + 2n^3 + 5 \\ &\leq 6n^4 + 2n^4 + 5n^4 \quad \text{when } n \geq 1 \\ &\leq 13n^4. \end{aligned}$$

So we set $N = 1$ and $c = 13$ in Definition 1.

Proof: $1 + 2 + \dots + n = O(n^2)$

Recall

$f(n) = O(g(n))$ if there exist constants $c > 0$ and N such that

$|f(n)| \leq c \cdot |g(n)|$ for all $n \geq N$.

Since $i \leq n$ for every $i = 1, \dots, n$,

$$1 + 2 + \dots + n = \sum_{i=1}^n i \leq \sum_{i=1}^n n = n \cdot n = n^2.$$

So we set $N = 1$ and $c = 1$ in Definition 1.

Proof: $1 + 2 + \dots + n = \frac{1}{2}n^2 + O(n)$

Recall that $1 + 2 + \dots + n = \frac{n(n+1)}{2} = \frac{1}{2}n^2 + \frac{1}{2}n$.

So

$$1 + 2 + \dots + n - \frac{1}{2}n^2 = \frac{1}{2}n.$$

Since $|\frac{1}{2}n| \leq \frac{1}{2} \cdot n$ for all $n \geq 1$, we set $N = 1$ and $c = \frac{1}{2}$ in Definition 1, giving

$$1 + 2 + \dots + n = \frac{1}{2}n^2 + O(n).$$

Proof: $1^2 + 2^2 + \dots + n^2 = \frac{1}{3}n^3 + O(n^2)$

Recall the sum of squares formula:

$$1^2 + 2^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6} = \frac{1}{3}n^3 + \frac{1}{2}n^2 + \frac{1}{6}n.$$

So

$$1^2 + 2^2 + \dots + n^2 - \frac{1}{3}n^3 = \frac{1}{2}n^2 + \frac{1}{6}n.$$

Since $n \leq n^2$ for $n \geq 1$, we get

$$\left| \frac{1}{2}n^2 + \frac{1}{6}n \right| \leq \frac{1}{2}n^2 + \frac{1}{6}n^2 = \frac{2}{3}n^2.$$

So we set $N = 1$ and $c = \frac{2}{3}$ in Definition 1, giving

$$1^2 + 2^2 + \dots + n^2 = \frac{1}{3}n^3 + O(n^2).$$

Proof: $\sum \frac{1}{i^2} = O(1)$

The trick: for $i \geq 2$,

$$\frac{1}{i^2} \leq \frac{1}{i(i-1)} = \frac{1}{i-1} - \frac{1}{i}.$$

Summing from $i = 2$ to n , this telescopes:

$$\sum_{i=2}^n \frac{1}{i^2} \leq \sum_{i=2}^n \left(\frac{1}{i-1} - \frac{1}{i} \right) = 1 - \frac{1}{n} < 1.$$

So

$$\frac{1}{1^2} + \sum_{i=2}^n \frac{1}{i^2} < 1 + 1 = 2.$$

We set $N = 1$ and $c = 2$ in Definition 1.

Proof: $\sum \frac{1}{i} = \ln n + O(1)$

Let us prove $\frac{1}{1} + \frac{1}{2} + \dots + \frac{1}{n} = \ln n + O(1)$, that is,

$$\frac{1}{1} + \frac{1}{2} + \dots + \frac{1}{n} - \ln n = O(1).$$

The idea is to approximate the sum by an integral.

“Think geometrically, prove algebraically.” — John Tate

Note that $\int_1^n \frac{1}{x} dx = \ln n$. Write

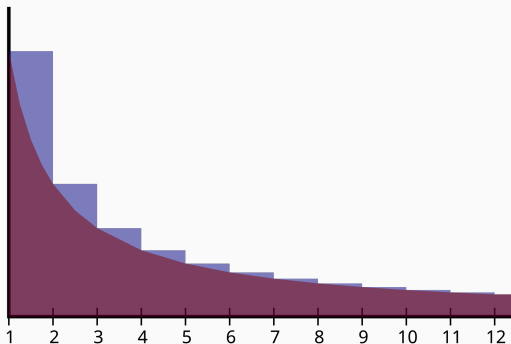
$$\ln n = \int_1^n \frac{1}{x} dx = \sum_{i=1}^{n-1} \int_i^{i+1} \frac{1}{x} dx < \frac{1}{1} + \frac{1}{2} + \dots + \frac{1}{n-1}.$$

So, $\frac{1}{1} + \frac{1}{2} + \dots + \frac{1}{n} - \ln n > 0$.

Proof: $\sum \frac{1}{i} = \ln n + O(1)$

Thus,

$$\begin{aligned} & \frac{1}{1} + \frac{1}{2} + \dots + \frac{1}{n} - \ln n \\ & \leq 1 + \sum_{i=1}^{n-1} \int_i^{i+1} \frac{1}{x} dx - \ln n \\ & = 1. \end{aligned}$$



Proof: $\sum \frac{1}{i} = \ln n + O(1)$

The limit

$$\frac{1}{1} + \frac{1}{2} + \cdots + \frac{1}{n} - \ln n$$

is the **Euler–Mascheroni constant**, approximately 0.577.

In many cases, it suffices to note that this expression is $O(1)$.

Write the following using big O notation.

1) $5 + 0.001n^3 + 0.025n$

2) $500n + 100n^{1.5} + 50n \log_{10} n$

3) $n^2 \log_2 n + n(\log_2 n)^2$

4) $n \log_3 n + n \log_2 n$

5) $3 \log_8 n + \log_2 \log_2 \log_2 n$

6) $n^{100} + 2^n$

Proposition 2

- 1) If $f_1 = O(g_1)$ and $f_2 = O(g_2)$, then $f_1 f_2 = O(g_1 g_2)$.
- 2) If $h = O(g)$, then $f \cdot h = O(f \cdot g)$.
- 3) If $f_1 = O(g_1)$ and $f_2 = O(g_2)$, then $f_1 + f_2 = O(\max(|g_1|, |g_2|))$. In particular, if $f_1 = O(g)$ and $f_2 = O(g)$, then $f_1 + f_2 = O(g)$.
- 4) If $f = O(g)$ and k is a constant, then $k \cdot f = O(g)$.

Definition 3

Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ and $g: \mathbb{R} \rightarrow \mathbb{R}$. We write $f(x, y) = O_y(g(x))$ if

$$(\forall y)(\exists c = c(y) > 0)(\exists N = N(y))(\forall x \geq N) (|f(x, y)| \leq c \cdot |g(x)|).$$

Examples

1) $m^2 \cdot 2^n = O_m(2^n)$

2) For $\delta > 0$,

$$\frac{1}{1^{1+\delta}} + \frac{1}{2^{1+\delta}} + \dots + \frac{1}{n^{1+\delta}} = O_\delta(1)$$

3) $1^k + 2^k + \dots + n^k = \frac{n^{k+1}}{k+1} + O_k(n^k)$

Proof: $m^2 \cdot 2^n = O_m(2^n)$

Let us prove, for fixed m , that $m^2 \cdot 2^n = O_m(2^n)$ by definition.

We need $c = c(m) > 0$ and $N = N(m)$ such that for all $n \geq N$,

$$|m^2 \cdot 2^n| \leq c \cdot |2^n|.$$

This holds immediately with $c = m^2$ and $N = 1$.

Proof: $\sum \frac{1}{j^{1+\delta}} = O_\delta(1)$

Let us prove, for $\delta > 0$,

$$\frac{1}{1^{1+\delta}} + \frac{1}{2^{1+\delta}} + \dots + \frac{1}{n^{1+\delta}} = O_\delta(1).$$

Applying the integral trick,

$$\begin{aligned} \text{L.H.S.} &= 1 + \sum_{i=2}^n \frac{1}{i^{1+\delta}} \\ &\leq 1 + \sum_{i=2}^n \int_{i-1}^i \frac{1}{x^{1+\delta}} dx \\ &= 1 + \int_1^n \frac{1}{x^{1+\delta}} dx \\ &= 1 + \left. \frac{x^{-\delta}}{-\delta} \right|_1^n \\ &\leq 1 + \frac{1}{\delta} \\ &= O_\delta(1). \end{aligned}$$

Proof: $1^k + \dots + n^k = \frac{n^{k+1}}{k+1} + O_k(n^k)$

Since x^k is increasing on $[0, \infty)$, for each $i \geq 1$,

$$\int_{i-1}^i x^k dx \leq i^k \leq \int_i^{i+1} x^k dx.$$

Summing over $i = 1, \dots, n$,

$$\int_0^n x^k dx \leq \sum_{i=1}^n i^k \leq \int_1^{n+1} x^k dx,$$

that is,

$$\frac{n^{k+1}}{k+1} \leq 1^k + 2^k + \dots + n^k \leq \frac{(n+1)^{k+1} - 1}{k+1}.$$

Proof: $1^k + \dots + n^k = \frac{n^{k+1}}{k+1} + O_k(n^k)$

It remains to bound the gap between the two sides. By the binomial theorem,

$$(n+1)^{k+1} = n^{k+1} + (k+1)n^k + \binom{k+1}{2}n^{k-1} + \dots + 1.$$

So

$$\frac{(n+1)^{k+1} - 1}{k+1} - \frac{n^{k+1}}{k+1} = n^k + \frac{1}{k+1} \left[\binom{k+1}{2}n^{k-1} + \dots \right].$$

Since k is fixed and every term after n^k has strictly lower degree, this whole expression is $O_k(n^k)$.

Combining with the sandwich on the previous slide,

$$1^k + 2^k + \dots + n^k = \frac{n^{k+1}}{k+1} + O_k(n^k).$$

Definition 4 ($\tilde{O}$ notation)

Let $f, g : \mathbb{R} \rightarrow \mathbb{R}$. We write $f(x) = \tilde{O}(g(x))$ if

$$(\exists c > 0)(\exists N)(\forall x \geq N) (|f(x)| \leq (\log x)^c \cdot |g(x)|).$$

In other words, $\tilde{O}$ hides polylogarithmic $(\log^{O(1)} n)$ terms.

Notation	Name
$O(1)$	constant
$O(\log n)$	logarithmic
$O(\log^c n)$	polylogarithmic
$O(n)$	linear
$O(n \log^c n)$ or $\tilde{O}(n)$	quasilinear
$O(n^2)$	quadratic
$O(n^c)$ or $n^{O(1)}$	polynomial
$2^{(\log n)^{O(1)}}$ or $n^{O(\log^c n)}$	quasipolynomial
$O(c^n)$ or $2^{O(n)}$	exponential

Definition 5 (Big-Omega notation)

Let $f, g : \mathbb{R} \rightarrow \mathbb{R}$. We write $f(x) = \Omega(g(x))$ if

$$(\exists c > 0)(\exists N)(\forall x \geq N)(|f(x)| \geq c \cdot |g(x)|).$$

Theorem 6 (Chernoff Bound)

Let $X_1, \dots, X_n$ be independent random variables with $X_i \in [0, 1]$. Let $S = X_1 + \dots + X_n$ and $\mu = \mathbb{E}[S]$. Then for any $\delta > 0$,

$$\Pr[S \geq (1 + \delta)\mu] \leq \exp\left(-\frac{\mu\delta^2}{3}\right) = \exp(\Omega(-\mu\delta^2)) = \exp(\Omega_\delta(-\mu)),$$

and for any $0 < \delta \leq 1$,

$$\Pr[S \leq (1 - \delta)\mu] \leq \exp\left(-\frac{\mu\delta^2}{2}\right) = \exp(\Omega(-\mu\delta^2)) = \exp(\Omega_\delta(-\mu)).$$

Definition 7 (Big-Theta notation)

Let $f, g : \mathbb{R} \rightarrow \mathbb{R}$ or $\mathbb{N} \rightarrow \mathbb{R}$. We write $f(x) = \Theta(g(x))$ if

$$(\exists c_1, c_2 > 0)(\exists N)(\forall x \geq N)(c_1 \cdot |g(x)| \leq |f(x)| \leq c_2 \cdot |g(x)|).$$

Definition 8 (Little-o notation)

Let $f, g : \mathbb{R} \rightarrow \mathbb{R}$. We write $f(x) = o(g(x))$ if

$$(\forall \epsilon > 0)(\exists N)(\forall x \geq N)(|f(x)| \leq \epsilon \cdot |g(x)|).$$

For example,

- $2n = o(n^2)$
- $\frac{1}{n}, \frac{1}{\log n}, \frac{1}{n^2}, \frac{1}{e^n}, \frac{1}{\log \log n} = o(1)$
- $\frac{1}{2^{100}} \neq o(1), \quad \frac{1}{2^{100}} = O(1)$

Definition 9 (Little-omega notation)

Let $f, g : \mathbb{R} \rightarrow \mathbb{R}$. We write $f(x) = \omega(g(x))$ if

$$(\forall c > 0)(\exists N)(\forall x \geq N)(|f(x)| > c \cdot |g(x)|).$$

Definition 10 (Little-theta notation)

Let $f, g : \mathbb{R} \rightarrow \mathbb{R}$. Write $f(x) = \theta(g(x))$ if

$$(\forall \epsilon > 0)(\exists N)(\forall x \geq N)(||f(x)| - |g(x)|| \leq \epsilon \cdot |g(x)|).$$

Equivalently, one may write $f \sim g$.

Example (Stirling's approximation):

$$n! \sim \sqrt{2\pi n} \left(\frac{n}{e}\right)^n$$

$$n! = \Theta \left(\sqrt{n} \left(\frac{n}{e}\right)^n \right)$$

Notation	Analogy
$f(n) = O(g(n))$	$\leq$
$f(n) = o(g(n))$	$<$
$f(n) = \Omega(g(n))$	$\geq$
$f(n) = \omega(g(n))$	$>$
$f(n) = \Theta(g(n))$	$\approx$
$f(n) = \theta(g(n))$	$=$